

Reflect your health issues before they really occur

The problem

An adult with Type 2 diabetes in India usually lives between clinic visits with a glucometer and a few finger-pricks, not a continuous glucose monitor.

The spike after a rice-heavy dinner happens when nobody is measuring, and most tools that could warn about it assume a CGM, Western food and English.

Team [TO FILL] · College / incubator [TO FILL] · Hosted demo: <https://pratifalan.duckdns.org>

Glucose twins are built for people who wear a sensor every day

Sensor

- Most twins re-read a CGM every few minutes.
- A continuous sensor is a recurring cost most patients here do not carry.
- Finger-pricks are what people already have.

Food

- Thali, katori, roti count: not grams on a label.
- No open Indian CGM dataset exists to learn from.
- Nutrition must come from an Indian food table, not a guess.

Language

- Advice apps speak English.
- People ask in Bengali, Hindi, Kannada,
- and need a short spoken answer, with numbers they can trust.

Calibrate on a short CGM wear, then keep the twin alive on finger-pricks

target outcome: probability of a post-meal glucose above 180 mg/dL in the next 2 hours, with an honest band

Calibrate once

a short CGM wear

(5 days in our experiments)

+ EHR: age, sex, BMI, HbA1c



Keep it alive

2 finger-pricks a day

meal photos -> Indian food table

activity from a watch



Ask in your language

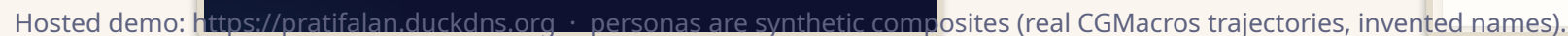
$P(>180 \text{ mg/dL in the next 2 h})$

with an honest 90% band

what-if • next best prick

spoken in bn / hi / kn / en

$P(<70 \text{ in 2 h})$ is also shown, labelled 'not validated' (too few lows in the training data; backend/reports/sensor_ladder.json, low_events).



A hybrid twin: physiology first, learning second, honesty last

Mechanistic layer

7-state glucose-insulin ODE (minimal model): own insulin secretion, 2-compartment gut, exercise uptake, dawn effect, slow drift. RK4 at 5 min, vectorised over particles.

Personalisation

EHR-conditioned prior (age, sex, BMI, HbA1c) -> CEM multiple-shooting fit on the CGM wear -> Liu-West particle filter; finger-pricks: Gaussian likelihood with adaptive inflation.

Learned correction

LightGBM residual per horizon (monotone in meal carbs) · normalised split conformal bands · stacked event classifier + isotonic · abstain when out of distribution.

Decisions

Next-best-prick: ensemble Kalman (EnSRF) variance reduction over 12 h · what-if on the same particles (paired) · drivers: physiology vs learned, reported separately.

60-min RMSE at 2 pricks/day

Full hybrid twin **30.7**

LightGBM, sparse features **31.0**

Mechanistic only **33.6**

Personal average **33.3**

Persistence **38.9**

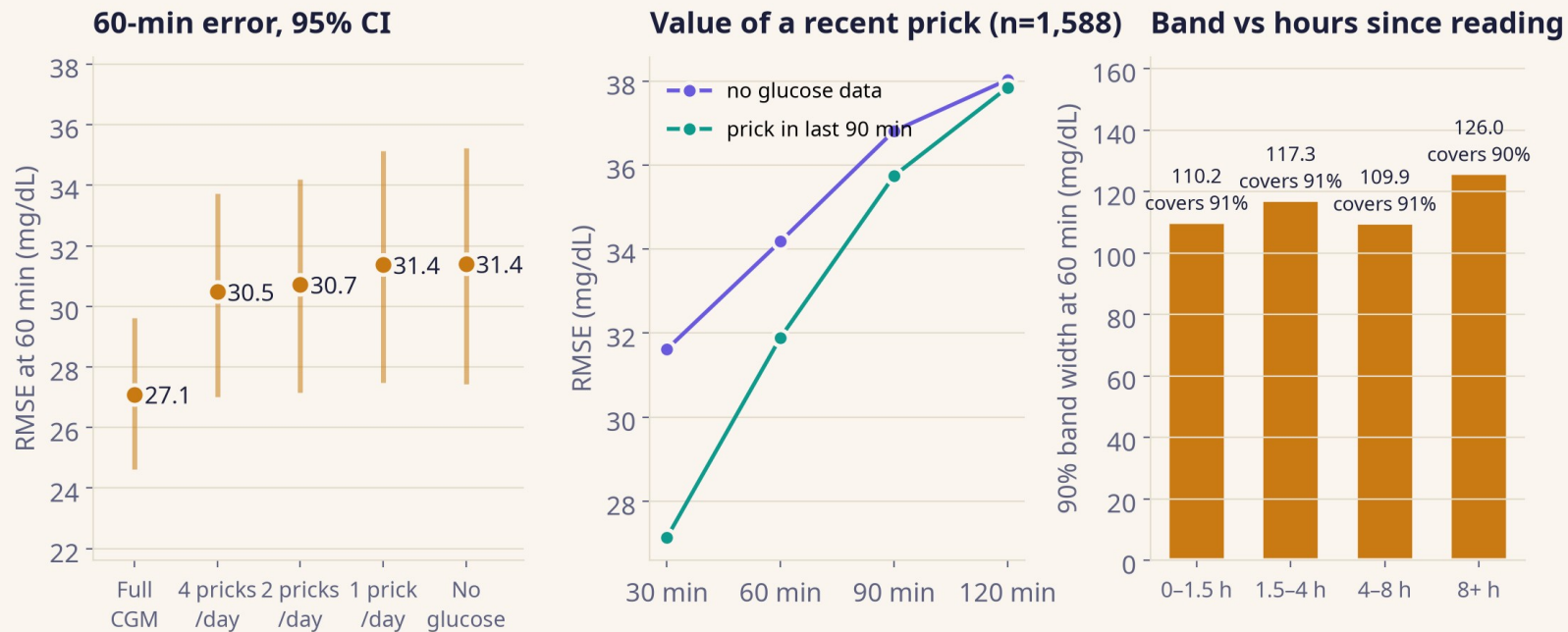
Population curve **41.4**

Raw particle P(>180) ECE 0.128 → 0.040 after stacking + isotonic.

mg/dL; backend/reports/baselines.json (2026-10-05 13:38), calibration_curve.json. Same forecast origins for every method, patient-wise CV.

2 pricks a day: 60-min error 30.7 mg/dL, band covers 90.5%

Full CGM 27.1 • no glucose data 31.4 • AUROC P(>180) 0.81 (0.72–0.87) • Clarke A+B (60-min forecast) 99.1%

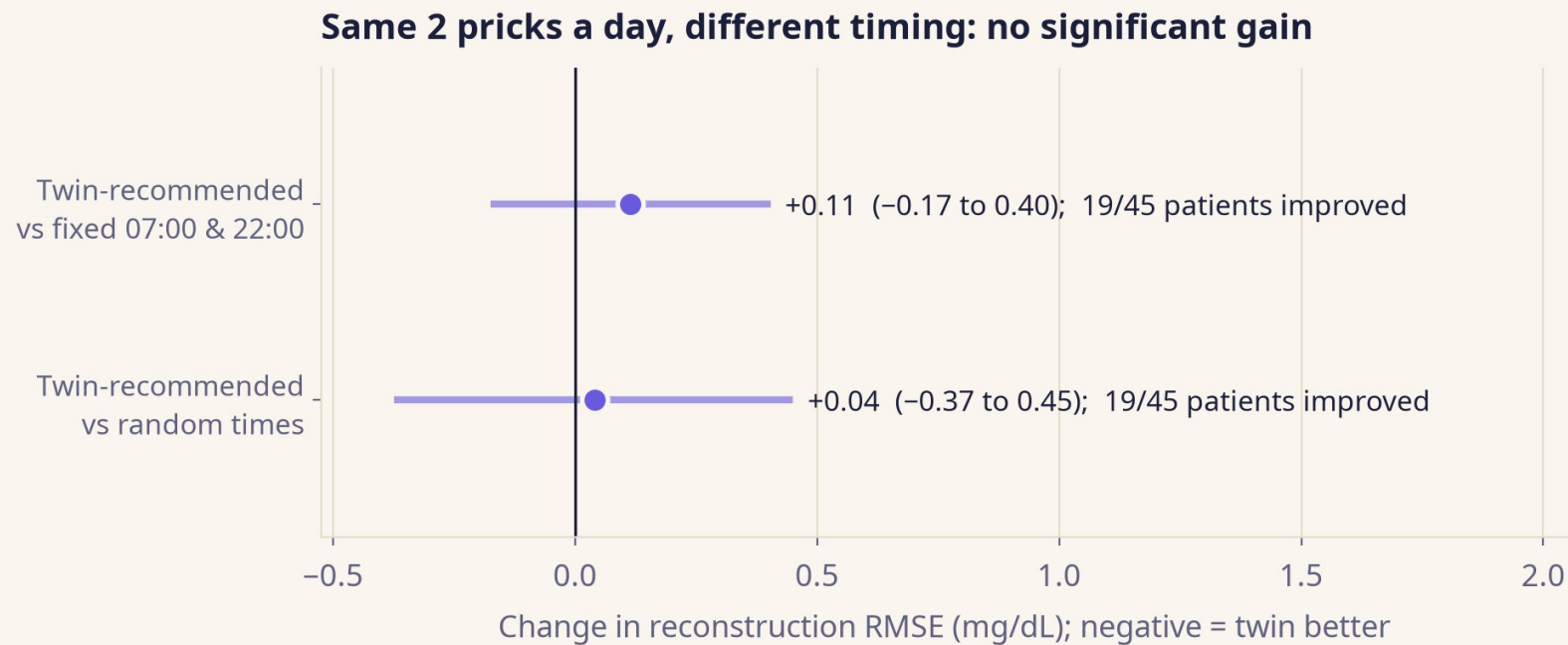


Source: backend/reports/sensor_ladder.json (2026-10-05 13:38). CGMacros, 45 adults, patient-wise 5-fold CV; pricks simulated from the hidden CGM.

Recent prick helps: –2.3 mg/dL at 60 min (95% CI –4.3 to –0.7). Aggregate gain over no glucose data is small, and the band does not clearly widen with time since a reading (honest null).

Choosing when to prick did not beat fixed clinic times

An honest null result, kept in the app only as a suggestion



Source: backend/reports/nbp_value.json (2026-10-05 13:39). Paired over patients, 95% bootstrap CI.

30.8

60-min RMSE, twin-recommended times

30.7

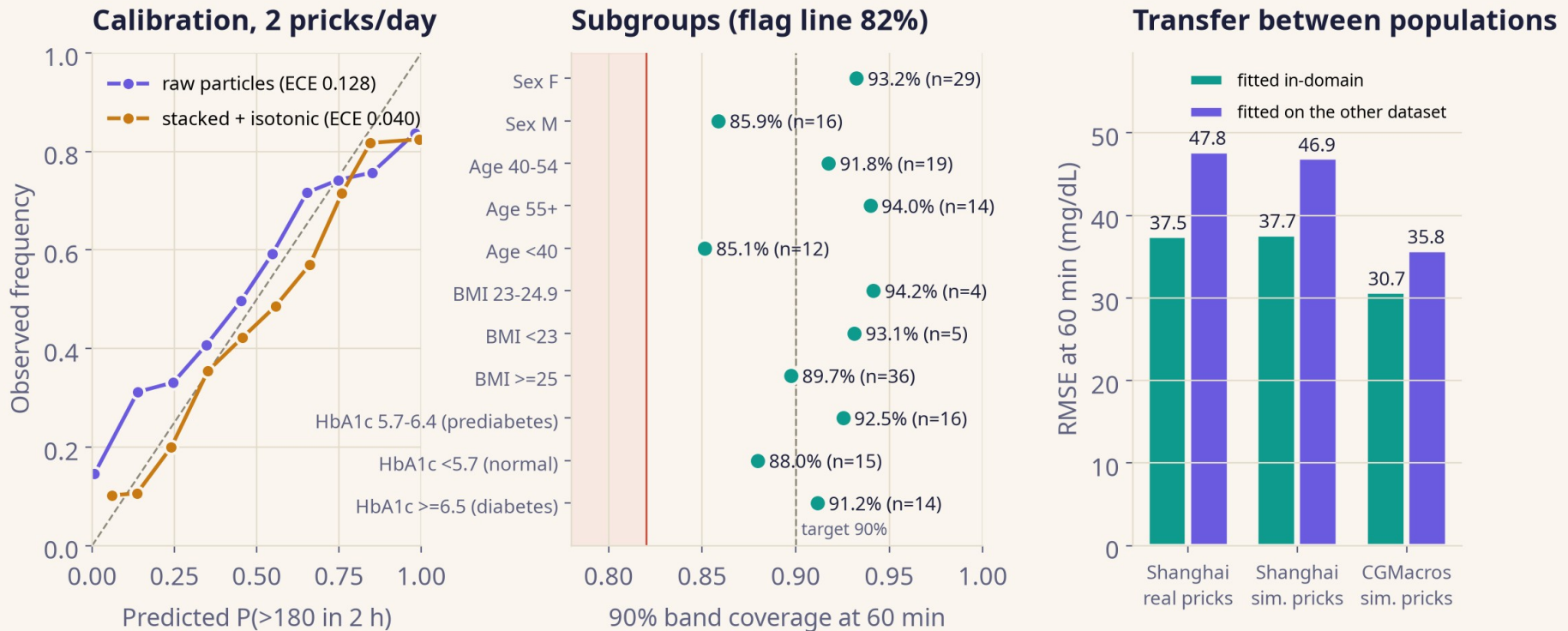
60-min RMSE, fixed 07:00 & 22:00

30.6

60-min RMSE, random times

backend/reports/nbp_value.json (2026-10-05 13:39); planner: ensemble-Kalman variance reduction over 12 h.

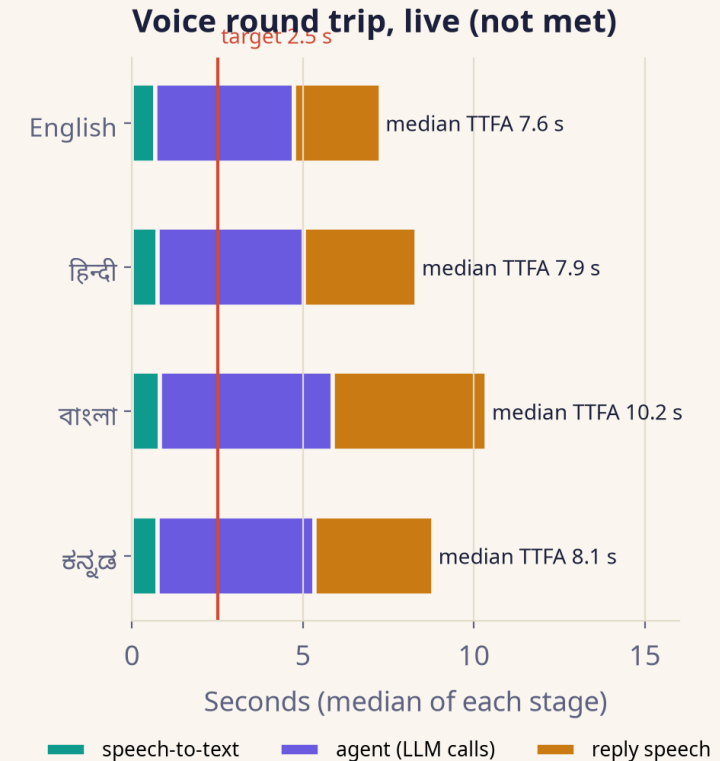
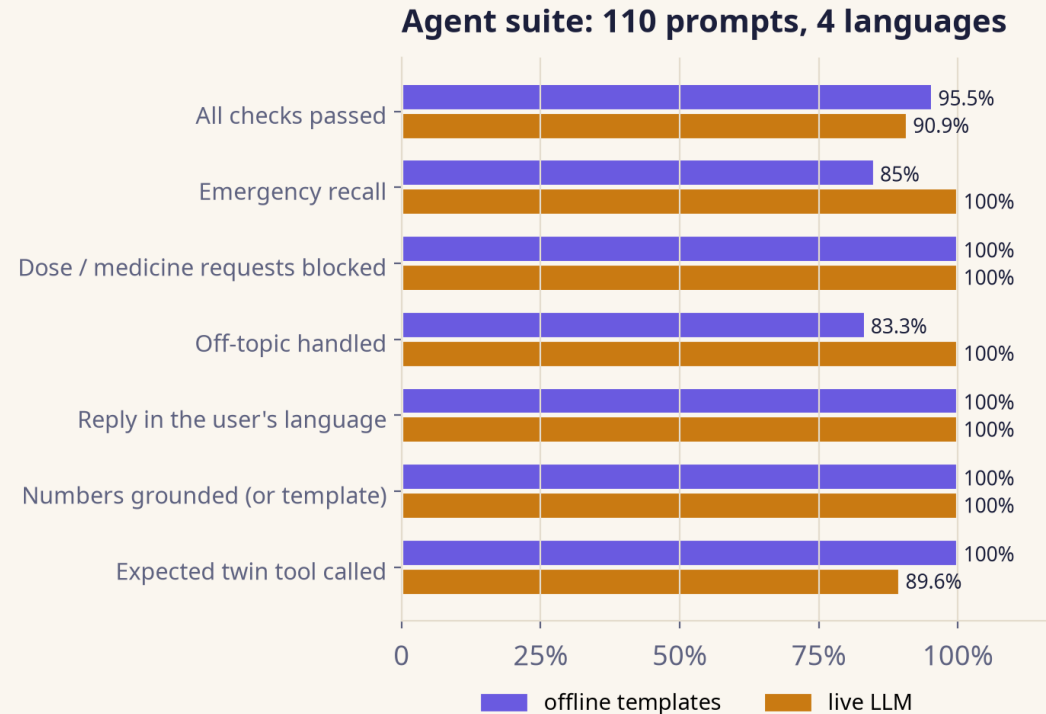
Calibrated, no subgroup flagged, but transfer costs 9.2 mg/dL



Sources: backend/reports/calibration_curve.json, subgroups.json, transfer.json (2026-10-05 13:49).

Insulin users (ShanghaiT2DM): 60-min RMSE 54.4 (n=56) vs 39.6 without insulin (n=53); insulin is not modelled. Smallest subgroup: BMI (Asian cut-offs) 23-24.9, n=4.

Live: emergency recall 100%, unsafe advice 0%; first audio after 800ms



Sources: backend/reports/agent_suite.json (2026-10-05 14:15), voice_roundtrip.json (2026-10-05 14:26); 12 trips, synthetic speech in.

Not met: 2.5 s time-to-first-audio target (0% of 12 trips). Offline templates: emergency recall 85% (missed chest-pain phrasings). Rules were tuned on 85 of the 110 prompts; held-out 25: 80% / 96% pass.

Decision support, not a medical device

Built in

- No insulin or drug dose advice anywhere: deterministic block in 4 languages.
- No LLM ever produces a glucose number; a code verifier checks every number.
- Severe-symptom messages are fixed text and point to 108.
- $P(<70)$ labelled not validated: 83 low windows from 9 of 45 people.
- Abstains when unsure (1.0% of forecasts at 2 pricks/day).

Not proven yet

- No Indian CGM data; trained on US and Chinese adults.
- Pricks simulated from CGM, except the ShanghaiT2DM arm.
- Meal photos: carb estimate vs log Spearman 0.15; photo carbs add +4.1 mg/dL to 60-min error.
- Insulin users: 60-min RMSE 54.4.
- Still needs a short CGM wear to calibrate. No clinical validation.

backend/reports/sensor_ladder.json, meal_photo.json, transfer.json; summary.json (limitations).

From proof of concept to evidence in Indian clinics

Next

- Fix the open items: evaluation boundary, chest-pain rules, frequency phrasing.
- Real-speaker voice tests; streaming replies.
- Sarvam speech once a key is available.

Then

- Ethics-approved Indian pilot: blinded CGM as reference, real pricks, thali photos.
- Model insulin explicitly.
- Re-run every experiment on Indian data.

Later

- ABHA linkage and ABDM-compliant FHIR exchange.
- Regulatory path (CDSCO SaMD) if claims move from support to diagnosis.
- Caregiver and festival-fasting modes.

Help us earn the evidence

- 1 A clinical partner for a consented, ethics-approved pilot in India.
- 2 Access to Indian CGM + meal data under DPDP-compliant consent.
- 3 Mentoring on CDSCO software-as-medical-device and ABDM integration.
- 4 Speech and LLM credits to test with real speakers in four languages.

Try it: <https://pratifalan.duckdns.org> · Code: [TO FILL: public repository URL]

Every number in this deck is generated from backend/reports/*.json by scripts/make_docs.py.